

**SUPPLEMENT B TO SF 252**  
**PROJECT 620-27-103**  
**(Replace Building 16 Roof – Phase 1)**  
**VA HVHCS – FDR – Montrose, NY**

Supplement B defines the minimum requirements of completion for each phase of the project design. This document applies to patient/administration area renovation and involves several design disciplines. The disciplines include but are not limited to Architectural, Electrical, Structural, Life Safety & Fire Protection design. In general, the project will consist of the following items in a hospital environment:

1. Landscape Architecture
2. Architecture,
3. NA
4. Architectural Specifications,
5. NA
6. NA
7. Structures,
8. Sanitary Systems,
9. NA
10. NA
11. NA
12. NA
13. Asbestos Abatement,
14. Lead Abatement,
15. Electrical,
16. Critical Path Method (CPM),
17. Value Engineering,
18. Estimating,
19. Barrier Free Design.
20. Fire Protection
21. NA

Primary design requirements include proficiency in the Architectural, Electrical, with secondary design expertise of the related systems necessary to accomplish the individual project items.

All designs will comply with the applicable VA standards and Federal Codes which include NFPA. In addition, all shall promote energy conservation incorporating LEED standard, ASHRAE Code and use of VA Sustainable Design Manual issued May 6, 2014 Rev1 August 18, 2017 as part of the design guide. Compliance with the use of VA Sustainable Manual is mandatory for all categories of construction, renovation, and non-recurring maintenance projects.

The AE Firm will submit a detailed schedule of tasks for individual disciplines within the Milestone Dates of the SP-13/ AE Design Schedule.

The A/E Firm shall conform to all design criteria, specification formats (MF 04) and drawing development in accordance with Technical Information Library (web), including but not limited to A/E Package TOC, PG-18-15 Minimum Requirements for A/E Submission, Design Review Checklists, Design Manuals, Specification, Cost Estimating and Commissioning. (<http://www.cfm.va.gov/TIL/index.asp>).

## REVIEW REQUIREMENTS

The following are minimum requirements for review purposes only. This does not relieve the Architectural/Engineering Firm of responsibility to produce a complete set of construction documents and estimates in accordance with industry standard practice and VA criteria.

- A. The Architectural/Engineering Firm shall develop and coordinate drawings, calculations, specifications and cost estimate. The degree of completion and the stages of submission shall be as specified.
- B. The Architectural/Engineering Firm shall submit documents, specification and / or drawings for review seven (7) Calendar days prior to each scheduled technical discipline review (TDR) meeting.
- C. For each TDR, Architectural/Engineering Firm shall report design development in accordance with submission and design criteria.
- D. At each TDR, the Architectural/Engineering Firm and technical disciplines shall review with VA red line mark-up comments and prepare technical discipline QA/QC report. A/E shall provide minutes of the meetings and distribute minutes and reports to attendees within five (5) Calendar days of the meeting.
- E. The Architectural/Engineering Firm shall submit Basis of Design Report, that includes but not limited to the following: Existing Condition & Field Survey, Photographs, Findings and Recommendations, Room Data Sheets, Field Report, Cost Estimate, Minutes of Meetings, TOC Specification, Design Checklist, Drawing & Document Register and Reports as requested by the COR at each

phase of the design submission.

- F. Provide computations and sizing calculations for Electrical, Structural, Fire Protection, and other specialty areas required for design development. For computerized calculations, submit complete and clear documentation of computer programs, interpretations of input/output and description of program procedures.
- G. NA
- H. The A/E Firm shall submit a construction cost estimate in RS Means format with the drawings at each project submission, prepared by an independent cost estimator. This estimate shall show the cost of construction which would be expected to be reflected by the construction contractors' bids, if the bids were submitted on the same day as the estimate. The level of detail for this estimate shall be consistent with the degree of completeness of the drawings being submitted with Divisional Breakdown and Additional Work and Cost differences reported from previous report. For detailed elements, "lump sum", "allowance" or "sq/ft" figures will not be accepted.
- I. Conduct periodic inspection services. These "site visits" include, but are not limited to, a pre-construction conference, intermediate and a final inspection visit (punch list). Depending on the complexity or length of the construction, more site visits may be required. The A/E Firm will submit a survey report summarizing the conditions found at the construction site as a result of each site visit, within five (5) Calendar days of each site visit.
- J. The A/E Firm will submit a survey report including photographs for each technical discipline summarizing the conditions found as a result of design development site visit(s), within five (5) business days of each site visit

- K. The Architectural/Engineering Firm will perform collection of as-built drawings and field verify floor plan, section, elevation, existing capacities and location of all utilities, design and upgrade as necessary to support the new renovation and extension. This may include field surveys to be performed as needed to verify existing record documents.

## PRELIMINARIES

- A. Preliminary drawings for construction projects are a means of identifying alternative approaches to correcting physical and functional deficiencies. The development of these drawings is an interactive process between VA staff and the Architectural/Engineering Firm in order to produce a narrative and graphical description of possible program alternatives. As appropriate, the VA shall make available prior to contract negotiations a tentative priority list of functional areas which may be included in the project, and a statement task.
- B. Preliminary Meeting: Prior to beginning any design work, a preliminary meeting shall be conducted between the Architectural/Engineering Firm and VA staff. At this stage, the scope of work shall be finalized and agreed to by all VA services involved in this project. At this point the Architectural/Engineering Firm shall have surveyed the existing facility to verify all existing conditions and become familiar with all deficiencies that need to be corrected. The Architectural/Engineering Firm shall write a summary of the meeting and submit it to the Contracting Officer for distribution and approval by the appropriate staff.
- C. The Architectural/Engineering Firm in collaboration with VA staff, shall illustrate in sketch form various possible program alternatives and refine them based on continuing interaction with the VA project team. The Architectural/Engineering Firm shall then complete and present alternative architectural solutions which are functionally and architecturally viable for consideration. The VA will select the final concept and subsequently identify the approved scope to be developed in the second preliminary (100%) review stage.

- D. The VA considers originality and imaginative design an essential part of the Architectural/Engineering Firm's responsibility. The comprehensive environmental approach is a necessity to all planning. Site, structures, vehicular and pedestrian circulation, open and screened areas and their relationships to each other must be combined to produce a plan that is functional and aesthetically pleasing.
- E. Complete conceptual alternatives must be displayed on the preliminary drawings. The work may include the division of an alternative into distinct parts, each of which has an identified priority. The preliminary drawings must, therefore, be developed so that irrespective of which parts (in priority order) are selected for further development in the Working Drawing phase. The solution shall provide a well designed facility with respect to functional operation, construction phasing, minimized disruption of existing hospital operation, and compatibility with structure and utilities between (1) existing facilities; (2) those for imminent construction; and, (3) those with low priority for future construction.
- F. A narrative to accompany the drawings shall identify construction phasing which shall be necessary to permit continuous operation of critical hospital operations. In addition, areas of new construction and renovation shall be shown. A detailed listing of submission requirements for phasing is included in a later paragraph.
- G. Subsequent to each VA review of the Architectural/Engineering Firm's work, the Architectural/Engineering Firm shall incorporate VA recommendations into the Preliminary Drawings.
- H. Technical Review Meetings (TDR): Review meetings shall be conducted at the end of each design phase. One (1) week prior to each scheduled TDR (set forth in SP-13 of the contract), submit five (5) copies of all narratives, drawings, calculations (including energy savings calculations) and cost estimates to be

reviewed by VA staff. The cost estimates shall reflect the level of completeness at each review stage. In addition, the Architectural/Engineering Firm shall take minutes of the TDR and submit two copies within five (5) Calendar days of the date of the TDR to the Contracting Officer for distribution to VA staff.



## 100% PRELIMINARY DESIGN

When specification sections are required, they shall meet the most current applicable standards and codes - federal, state and local, at the time the Contract is awarded.

### A. LANDSCAPE ARCHITECTURAL

- 1) All plans are to be presented on VA standard size drawings and submitted both on paper and in native AutoCAD format, the latest version. The drawings must delineate new construction from existing conditions. The Architectural/Engineering firm shall conduct a site survey and review all historical drawings located at Hudson Valley Veterans Health Care campuses pertaining to the project (utilities, landscaping, etc) prior to commencing design work. Architectural/Engineering Firm shall provide 1 or more design alternatives where feasible. Those areas of the design that do not have cost effective alternatives shall not require a design alternative. Alternatives designs shall be chosen during the design review of this phase for further development in subsequent submissions.

### B. ARCHITECTURAL

1. The drawings required by this submittal delineate the one approved scheme from the previous submittal. Submit a layout for all floors and roof areas. These drawings shall incorporate all revisions from comments on the previous submittal and should show all rooms, doors, corridors, basic column grid, column sizes, electrical closets, phone closets, equipment rooms, mechanical shafts, space and all vertical circulation.

2. Show preliminary building sections to define building configuration, area and volume. Typical wall sections showing proposed building systems and materials shall be drawn.
1. Using the floor plans, clearly define smoke compartments and exits from each compartment. Also indicate sprinklered areas (if any), location of fire extinguisher cabinets, exit paths and distance to the stair and employee occupancy.

#### C. INTERIOR DESIGN

No requirement at this submission.

#### D. ARCHITECTURAL SPECIFICATIONS

1. Submit the technical architectural specification sections of VA Master Construction Specifications edited to reflect the scope of work of the project. Coordinate drawings with all changes to the specifications.
2. All specification sections must be prepared in a professional manner. No writer's notes or // slashes shall be included in the specification sections. In addition, all required references or reference specification sections shall be clearly identified in each specification section and red line changes identified for VA approval

#### E. EQUIPMENT SPECIFICATIONS

No requirement at this submission.

## F. STRUCTURAL

1. No requirement at this submission.

## G. SANITARY

Submit design narrative prepared in the first review that has been adjusted for the site scheme selected. Include corrections and changes.

2. Using the site scheme selected in the first review, submit a preliminary utility plot plan showing the existing and proposed trunk sewers and the water distribution system in the entire facility. Include locations of connections to existing VA owned utility systems.

## H. PLUMBING

NA

## I. HEATING, VENTILATION AND AIR CONDITIONING (HVAC)

NA

## J. ASBESTOS/PCB ABATEMENT

1. Prior to this submission of the final exposure assessment report, the CIH shall have completed the following work:
  - a. Tasks 1 through 4 according to the "Special Instructions – Asbestos/PCB Abatement" document (attached)

2. Submit an exposure assessment report on the impacted areas of the project. Include the following information:
  - a. Describe the impacted areas by sketches, room names or other means. Show or describe the location of each sample taken, its composition and identification number.
  - b. Classify each impacted space as "Major Abatement", "Minor Abatement" or "No Abatement" areas. A "Major Abatement" area requires full protection measures (sealing off, change room, shower room, etc.), and removal of all asbestos within the limits of the area. A "Minor Abatement" area can be decontaminated with one or more applications of the "glove and bag" approach. A "No Abatement" area requires no abatement work.
  - c. For each "Major Abatement" area, identify area location within building, and provide quantity and cost estimates of all asbestos within the limits of the area. Quantities are to be in linear feet of pipe and square feet of ceilings and walls. The cost is to be derived from the quantities multiplied by a unit price.
  - d. For each "Minor Abatement" area, provide the description, size and location of each pipe element to be decontaminated and include a cost estimate.
  - e. Provide a summary of the square footage of floor space for all major and minor abatement areas, and the total linear and square foot of asbestos to be abated, the total cost of abatement and a preliminary schedule of completion. Include in the cost estimate, any cost for

decontamination of equipment and fixtures. Provide a copy of the above summary to the construction cost estimator for inclusion as a separate bid item in the project estimate.

- f. Provide an assessment on the impact of abatement on building sub-systems such as ceilings, lights, walls, carpets, pipes and equipment. Provide room by room description and estimated quantities of impact items to be replaced (number of lighting fixtures, square feet of ceiling, etc.) and a summary of these quantities. Provide a cost estimate for all impacted items.
- g. Provide a signed cover statement.

#### K. LEAD ABATEMENT

- 1. Prior to this submission of the final exposure assessment report, the CIH shall have completed the following work:
  - a. Tasks 1 through 4 according to the "Special Instructions – LEAD Abatement" document (attached)
- 2. Submit an exposure assessment report on the impacted areas of the project. Include the following information:
  - a. Describe the impacted areas by sketches, room names or other means. Show or describe the location of each sample taken, its composition and identification number.

- b. Classify each impacted space as "Major Abatement", "Minor Abatement" or "No Abatement" areas. A "Major Abatement" area requires full protection measures (sealing off, change room, shower room, etc.), and removal of all asbestos within the limits of the area. A "Minor Abatement" area can be decontaminated with one or more applications of the "glove and bag" approach. A "No Abatement" area requires no abatement work.
- c. For each "Major Abatement" area, identify area location within building, and provide quantity and cost estimates of all LEAD within the limits of the area. Quantities are to be in linear feet of pipe and square foot of ceilings and walls. The cost is to be derived from the quantities multiplied by a unit price.
- d. For each "Minor Abatement" area, provide the description, size and location of each pipe element to be decontaminated, **and** include a cost estimate.
- e. Provide a summary of the square footage of floor space for all major and minor abatement areas, the total linear and square foot of LEAD to be abated, the total cost of abatement and a preliminary schedule of completion. Include in the cost estimate, any cost for decontamination of equipment and fixtures. Provide a copy of the above summary to the construction cost estimator for inclusion as a separate bid item in the project estimate.
- f. Provide an assessment on the impact of abatement on building sub-systems such as ceilings, lights, walls, carpets, pipes and equipment. Provide room by room description and estimated quantities of impact

items to be replaced (number of lighting fixtures, square feet of ceiling, etc.) and a summary of these quantities. Provide a cost estimate for all impacted items.

- g. Provide a signed cover statement.

#### L. ELECTRICAL

1. Provide a statement of the impact of the new construction to the existing distribution system. Include the projected load of the new construction. The narrative must include the advantages/disadvantages to support the systems as recommended by the Engineering Firm.
2. Provide a sheet for symbols, lighting fixture schedules, notes and abbreviations used at this submission.
3. Submit a clear, concise elementary one-line diagram of the proposed electrical system including all new equipment.
4. Provide a plot plan of the existing electrical distribution system.

#### M. BARRIER FREE DESIGN

1. No requirement at this submission.

#### N. FIRE PROTECTION

1. No requirement at this submission.

AE shall submit two (2) full size sets and one (1) half size set of hard copies design drawings and specifications including an electronic copy in PDF format. Also, all electronic copies including CAD format drawings will be provided to the VA via mandated Autodesk Build File Management system.



## WORKING DRAWINGS

1. The final approved preliminary plan shall be the basis for the development of the Working Drawings. Any changes from this plan that the Architectural/Engineering Firm proposes must be approved by the VA Project Manager prior to proceeding with the Working Drawings.
2. All plans shall be reviewed for functional and aesthetic relationships. The result of this phase shall be a set of design documents defined to the point that no further functional decisions are required.
3. Some of the project elements may not be covered by a VA Master Specification Section. It is the responsibility of the Architectural/Engineering Firm to develop the final specification requirements for each unit of work introduced into the project. New sections developed by the Architectural/Engineering Firm shall be in the same format as that used in the VA Master Specifications.
4. Review meetings shall be conducted at the end of each working drawing design phase. Two (2) weeks prior to each scheduled review meeting (set forth in SP-13 / AE Design Schedule of the contract), submit Three (3) copies of drawings, specifications and cost estimates for review. The cost estimate shall reflect the level of completeness of each stage. After final approval of contract drawings has been received and the packages is ready for bid, submit 3 sets of specification and drawing documents and one set of drawings, specifications, and cost estimates on Autodesk build. Drawing files shall be in AutoCAD format, the latest version. A multipage DWF file containing all drawings in the package or shared drive access shall also be

provided. Specification documents shall be in Microsoft Word format.  
Estimate file shall be in Microsoft Excel format.

## 65% WORKING DRAWINGS

When specification sections are required, they shall meet the most current applicable standards and codes - federal, state and local, at the time the Contract is awarded.

### A. LANDSCAPE ARCHITECTURAL

- 1) All plans are to be presented on VA standard size drawings and submitted both on paper and in native AutoCAD format, the latest version. The drawings must delineate new construction from existing conditions.
- 2) Submit applicable sections of the VA Master Specifications (MF2004 edition) edited to reflect the scope of work of the project.
- 3) Submit plans with all of the revisions from reviews in the Preliminary Phase incorporated into the plans.
- 4) Include any manufacturers cut sheets for any new buildings or proposed structures.
- 5) Include any required elevations and section view drawings.

### B. ARCHITECTURAL

1. Submit plans with all of the revisions required from reviews in the Preliminary Phase incorporated into the plans.
2. Submit floor plans for each floor, new as well as renovated, showing all plumbing fixtures, door locations and swing, smoke and fire rated partitions

and fire extinguisher cabinets. Show wall thickness and chase walls. All plans shall show column grid with columns indicated and expansion joints. Indicate and identify fixed equipment. Identify wheelchair accessible facilities.

3. Submit reflected ceiling plans for the entire scope indicating ceiling mounted equipment, lighting fixtures, air diffuser registers, etc.
4. Submit equipment floor plans incorporating comments from previous review meetings. All existing and new equipment to be installed must be identified.
5. Building sections, as needed, to define building configuration. All finish floor elevations must be indicated, as well as interstitial space, if applicable.
6. Submit typical wall sections showing construction, building materials and systems, and proposed sill and head heights of openings.
7. Show drafting symbols and abbreviations per VA standards, general notes and begin drawings for partition, door, window and louver schedules.
8. Submit demolition plans, existing finish schedule and notes, on demolition plan.
9. Provide a list of provision of LEED, when required.

#### C. INTERIOR DESIGN

1. NA

#### D. ARCHITECTURAL SPECIFICATIONS

1. Assure the original specification drafts have been edited and tailored in their application to represent accurate coordination between the drawings and specifications.
2. All specification sections must be prepared in a professional manner. No writer's notes or // slashes shall be included in the specification sections. In addition, all required references or reference specification sections shall be clearly identified in each specification section and red line changes identified for VA approval.
3. Submit one full set of drawings of all disciplines, fully coordinated.

#### E. EQUIPMENT SPECIFICATIONS

NA

#### F. STRUCTURAL

1. Submit applicable sections of the VA Master Specifications (MF2004 edition) edited to reflect the scope of work of the project.
2. Drawings and edited specification documents shall not contradict each other.
3. Show necessary typical details correlating architectural and mechanical features.
4. Show sections and details so that construction features can be defined, including the interfacing with new to existing construction.

#### G) Hazardous Evaluation Form (SOP 138-022)

Meet, review and execute the Hazardous Evaluation Form (SOP 138-022)  
Section A and C for the referenced project at 65 % and 95% working drawing  
submission review

#### H. SANITARY

1. Add building service piping to preliminary utility plans. Include approximate size of all existing pipe and invert elevations of trunk sewers.
2. Submit calculations.
1. Submit sections of the VA Master Specifications and Details edited to reflect the scope of work of the project. Coordinate drawings with all changes to the specifications. All specification sections must be prepared in a professional manner. No writer's notes or // slashes shall be included in the specification sections. In addition, all required references or reference specification sections shall be clearly identified in each specification section and red line changes identified for VA approval.

#### I. PLUMBING

1. Submit applicable sections of the VA Master Specifications (MF2004 edition) edited to reflect the scope of work of the project.
2. Drawings and edited specification documents shall not contradict each other.
3. Submit all calculations.
4. Provide complete selection and sizing data, including catalog cuts and calculations.
5. Submit any applicable equipment schedules.

## J. HEATING, VENTILATION AND AIR CONDITIONING (HVAC)

NA

## K. ASBESTOS/PCB ABATEMENT

1. Prepare drawings and specifications for the asbestos and PCB abatement work reflecting the exposure assessment report. The drawings shall be separately identified from the rest of the drawings, and shall provide as a minimum the following information:
  - a. For each "Major Abatement" area show the limits of sealing of, the location and quantities of asbestos material, the arrangements for auxiliary rooms, such as change room, shower room, etc., the engineering of the negative air system, the path of asbestos transportation to the loading platform and location and connection to required utilities.
  - b. For each "Minor Abatement" area show on plans the exact location type and length of pipe element to be abated by the "Glove and Bag" approach and any other abatement features.
  - c. Provide a major and minor summary of the square footage of floor space for all abatement areas, the total linear and square foot of asbestos to be abated and the total cost of abatement. Include in the cost estimate any cost for decontamination of equipment and fixtures. Provide a copy of the above summary to the construction

cost estimator for inclusion as a separate bid item in the project estimate.

#### L. LEAD ABATEMENT

1. Prepare drawings and specifications for the LEAD abatement work reflecting the exposure assessment report. The drawings shall be separately identified from the rest of the drawings, and shall provide as a minimum the following information:
  - a. For each "Major Abatement" area show the limits of sealing of, the location and quantities of asbestos material, the arrangements for auxiliary rooms, such as change room, shower room, etc., the engineering of the negative air system, the path of asbestos transportation to the loading platform and location and connection to required utilities.
  - b. For each "Minor Abatement" area show on plans the exact location type and length of pipe element to be abated by the "Glove and Bag" approach and any other abatement features.
  - c. Provide a major and minor summary of the square footage of floor space for all abatement areas, the total linear and square foot of asbestos to be abated and the total cost of abatement. Include in the cost estimate any cost for decontamination of equipment and fixtures. Provide a copy of the above summary to the construction cost estimator for inclusion as a separate bid item in the project estimate.

## M. ELECTRICAL

1. Submit applicable sections of the VA Master Specifications (MF2004 edition) edited to reflect the scope of work of the project.
2. Submit all calculations
3. Provide complete selection and sizing data, including catalog cuts and calculations.
4. Submit any applicable equipment schedule.
5. Provide a sheet for symbols, notes and abbreviations used in this submission.
6. Submit a full set of electrical, lighting, power and signal floor plans.
7. Provide demolition plans to include the complete electrical work in all areas to be renovated.
8. Submit sections of the VA Master Specifications and Details edited to reflect the scope of work of the project. Coordinate drawings with all changes to the specifications.

## N. BARRIER FREE DESIGN

4. Submit plans incorporating comments from the previous review.

## O. FIRE PROTECTION

1. Submit plans showing layout of new/relocated equipment incorporating comments from the previous review.

## P. 138-022 Form

1. Complete and sign off 138-022 Form



AE shall submit two (2) full size sets and one (1) half size set of hard copies design drawings and specifications including an electronic copy in PDF format. Also, all electronic copies including CAD format drawings will be provided to the VA via mandated Autodesk Build File Management system.

AE shall also submit a copy of comment resolution addressing fire protection issues to the VA for review.

## 95% WORKING DRAWINGS

When specification sections are required, they shall meet the most current applicable standards and codes - federal, state and local, at the time the Contract is awarded.

1. The 95% Working Drawing phase involves the production of complete drawings, specifications, and other documents necessary for the bidding and construction of the project. Also included at this phase are the final detailed cost estimate, the final phasing plan and the construction schedule.
2. It is the Architectural/Engineering Firm's responsibility to provide a quality set of documents. Related documents shall be complete, fully coordinated and ready for reproduction for contract.
3. Prior to reproduction for issue for construction bids, the Architectural/Engineering Firm shall make any changes to the documents identified as necessary.
4. Submit a complete set of applicable VA Master Specifications edited to reflect the scope of work of the project. Also, submit copies of any architectural specifications prepared by the Architectural/Engineering Firm from any source other than VA Master Specifications. Ensure the specifications have been edited and tailored in their application to represent accurate coordination between the drawings and specifications.

### A. LANDSCAPE ARCHITECTURAL

1. All plans are to be presented on VA standard size drawings and submitted both on paper and in native AutoCAD format, the latest version. The drawings must delineate new construction from existing conditions.
2. Submit plans with all of the revisions from reviews in the Preliminary Phase incorporated into the plans.
3. Include any manufacturers cut sheets for any new buildings or proposed structures.
4. Include any required elevations and section view drawings.

## B. ARCHITECTURAL

1. Submit a 95% complete and coordinated set of construction drawings and specifications.
2. Submit fully dimensioned floor plans showing all revisions required by comments from the previous review meetings.
3. Submit interior details.
4. Submit drafting symbols, and abbreviations, general notes and schedules that are complete and coordinated with all contract documents.
5. Submit a complete and coordinated finish schedule.
6. Submit demolition plans, existing finish schedule and notes, on demolition plans.
7. Submit completed building sections, wall sections and exterior elevations that show finish floor elevations, and indicate all building systems and materials.

8. Submit completed reflected ceiling plans for entire buildings, indicating all ceiling mounted equipment, lighting fixtures, air diffusers, registers, etc.
9. Submit equipment plans and details with all revisions from comments on previous submittal.
10. Provide a list of provision of LEED meeting the VA Energy Conservation requirements, when required.

#### C. INTERIOR DESIGN

1. NA

#### D. ARCHITECTURAL SPECIFICATIONS

1. Assure the original specification drafts have been edited and tailored in their application to represent accurate coordination between the drawings and specifications.
2. All specification sections must be prepared in a professional manner. No writer's notes or // slashes shall be included in the specification sections. In addition, all required references or reference specification sections shall be clearly identified in each specification section and red line changes identified for VA approval
2. Submit one full set of drawings of all disciplines, fully coordinate.
3. Submit a brief description of work for inclusion in the Pre-Solicitation Notice.

4. Final Bid Document Submittal:

a. Revise draft specifications to incorporate:

1. All changes, resolution of conflicts and modifications noted in previous reviews.
2. Results of any drawing changes not shown on the drawings that affect the specifications.

b. Type the specifications in final format and content including any desk copy changes made by the VA staff at the previous reviews. Submit a complete set of the typed architectural specifications including one full set of final drawings of all disciplines, fully coordinate.

E. EQUIPMENT SPECIFICATIONS

NA

F. STRUCTURAL

1. Submit applicable sections of the VA Master Specifications (MF2004 edition) edited to reflect the scope of work of the project.
2. Drawings and edited specification documents shall not contradict each other.
3. Submit all calculations.
4. Show necessary typical details correlating architectural and mechanical features.

5. Show sections and details so that construction features can be defined, including the interfacing with new to existing construction.

G. Hazardous Evaluation Form (SOP 138-022)

Meet, review and execute the Hazardous Evaluation Form (SOP 138-022) Section A and C for the referenced project at 65 % and 95% working drawing submission review

H. SANITARY

1. Submit 95% complete drawings to include the following:
  - a. Previously submitted drawings that have incorporated comments of the last review.
  - b. Legend, notes, and details.

I. PLUMBING

1. Submit applicable sections of the VA Master Specifications (MF2004 edition) edited to reflect the scope of work of the project.
2. Drawings and edited specification documents shall not contradict each other.
3. Submit all calculations
4. Provide complete selection and sizing data, including catalog cuts and calculations
5. Submit any applicable equipment schedules

## J. HEATING, VENTILATION AND AIR CONDITIONING (HVAC)

NA

## K. ASBESTOS/PCB ABATEMENT

1. Provide details, sections, notes and other information that is necessary for complete bid documents.
2. Prepare project specifications by editing and modifying VA Master Specifications on asbestos and PCB abatement.
3. All specification sections must be prepared in a professional manner. No writer's notes or // slashes shall be included in the specification sections. In addition, all required references or reference specification sections shall be clearly identified in each specification section and red line changes identified for VA approval
3. Provide the drawings and specifications for the restoration of the impacted building sub-systems as an integral part of the overall project design. Provide for replacing of removed asbestos insulation with acceptable non-asbestos insulation, replacing contaminated carpets, replacing old contaminated light fixtures and replacing partially torn down plaster ceilings. Coordinate work with the Architectural/Engineering Firm.
4. Provide an integrated phasing plan showing how the abatement and the general modification work will be executed. Provide some reasonable cushion between each abatement phase and the following general construction phase to allow for unforeseen delays in abatement completion.

5. Provide a major and minor summary of the square feet of floor space for all abatement areas, the total linear and square feet of asbestos to be abated and the total cost of abatement. Include in the cost estimate any cost for decontamination of equipment and fixtures. Provide a copy of the above summary to the construction cost estimator for inclusion as a separate bid item in the project estimate.
6. Perform Task 5 according to the "Special Instructions – Asbestos/PCB Abatement" document (attached)

#### L. LEAD ABATEMENT

1. Provide details, sections, notes and other information that is necessary for complete bid documents.
2. Prepare project specifications by editing and modifying VA Master Specifications on asbestos abatement.
3. All specification sections must be prepared in a professional manner. No writer's notes or // slashes shall be included in the specification sections. In addition, all required references or reference specification sections shall be clearly identified in each specification section and red line changes identified for VA approval
4. Provide the drawings and specifications for the restoration of the impacted building sub-systems as an integral part of the overall project design. Provide for replacing of removed asbestos insulation with acceptable non-asbestos insulation, replacing contaminated carpets, replacing old,



contaminated light fixtures and replacing partially torn down plaster ceilings. Coordinate work with the Architectural/Engineering Firm.

5. Provide an integrated phasing plan showing how the abatement and the general modification work will be executed. Provide some reasonable cushion between each abatement phase and the following general construction phase to allow for unforeseen delays in abatement completion.
6. Provide a major and minor summary of the square feet of floor space for all abatement areas, the total linear and square feet of LEAD to be abated and the total cost of abatement. Include in the cost estimate any cost for decontamination of equipment and fixtures. Provide a copy of the above summary to the construction cost estimator for inclusion as a separate bid item in the project estimate.
7. Perform Task 5 according to the "Special Instructions – LEAD Abatement" document (attached)

#### M. ELECTRICAL

1. Submit applicable sections of the VA Master Specifications (MF2004 edition) edited to reflect the scope of work of the project.
2. Drawings and edited specification documents shall not contradict each other.
3. Submit all calculations
4. Provide complete selection and sizing data, including catalog cuts and calculations.
5. Submit any applicable equipment schedules.

6. Provide a sheet for symbols, notes and abbreviations used at this submission.
7. Submit a full set of electrical, lighting, power, and signal floor plans.
8. Provide demolition plans to indicate the complete electrical work in all areas to be renovated.
9. Phasing and shutdown requirements.

#### N. BARRIER FREE DESIGN

1. Complete all drawings and specifications. Ensure that every effort to make this facility accessible to handicapped employees and patients has been taken in this design.

#### O. FIRE PROTECTION

1. Complete all plans and specifications. Ensure compliance with NFPA and other applicable codes.

#### P. Estimating

8. Engineering firm will provide a complete and accurate cost estimate.
9. Construction estimate will be in R.S. Means format.

#### Q. PHASING REQUIREMENTS

1. Phasing requirements shall describe the general sequence of the project work, estimated project duration (including allowances for delivery items),

and what government constraints shall exist that will influence the Contractor's approach to the construction project. In addition, special attention shall be given to asbestos abatement requirements to ensure that the project phasing plan and associated cost are reasonable. Adequate time shall be allocated for the evaluation for asbestos abatement areas in addition to the time to perform the asbestos abatement work that must precede the general construction. The Architectural/Engineering Firm shall submit the following phasing information:

- a. Phasing Narrative in written form which outlines phasing requirements and sequence with all areas of the project identified as a part of some phase. Each phase description shall include constraints particular to that phase, what other phases that must precede it, and any VA moves which must precede the start of the phase or phases. If equipment and other removable items require storage and relocation by the government, because of asbestos abatement, these requirements shall be listed in the phasing narrative. Special phasing constraints which may be common to the project should be listed at the end of the narrative and not within each individual phase description.
  - b. Individual phases shall be outlined and labeled on all drawings including site, architectural, Structural, plumbing and electrical drawings. Phases shall be outlined on the submitted full sized drawings.
- 
- a. All systems shall be designed so that, upon completion of a particular phase, the entire area covered by that phase can be occupied by the Medical Center personnel with all systems functioning properly.

## R. ESTIMATING

1. Engineering firm will provide a complete and accurate cost estimate.
2. Construction estimate will be in R.S. Means format

## S. Meetings

1. Meet, review and execute the Hazardous Evaluation Form (SOP 138-022) Section A and C for the referenced project at 65 % and 95% working drawing submission review.

AE shall submit two (2) full size sets and one (1) half size set of hard copies design drawings and specifications including an electronic copy in PDF format. Also, all electronic copies including CAD format drawings will be provided to the VA via mandated Autodesk Build File Management system.

AE shall also submit a copy of the comment resolution addressing fire protection issues to the VA for review.

## 100% WORKING DRAWINGS

When specification sections are required, they shall meet the most current applicable standards and codes - federal, state and local, at the time the Contract is awarded.

1. The 100% Working Drawing phase involves the production of complete drawings, specifications, and other documents necessary for the bidding and

construction of the project. Also included at this phase are the final detailed cost estimate, the final phasing plan and the construction schedule.

2. It is the Architectural/Engineering Firm's responsibility to provide a quality set of documents. Related documents shall be complete, fully coordinated and ready for reproduction for contract.
3. Prior to reproduction for issue for construction bids, the Architectural/Engineering Firm shall make any changes to the documents identified as necessary.
4. The Architectural/Engineering Firm shall deliver the original contract drawings to the Project Manager for signature, after the previous review comments are incorporated and the contract drawings have been approved. The contract drawings shall bear the seal of the Registered Architect and Professional Engineer responsible for the design.
5. Submit a complete set of applicable VA Master Specifications edited to reflect the scope of work of the project. Also, submit copies of any architectural specifications prepared by the Engineering Firm from any source other than VA Master Specifications. Ensure the specifications have been edited and tailored in their application to represent accurate coordination between the drawings and specifications.

#### A. LANDSCAPE ARCHITECTURAL

1. All plans are to be presented on VA standard size drawings and submitted both on paper and in native AutoCAD, the latest version. The drawings must delineate new construction from existing conditions.

2. Submit applicable sections of the VA Master Specifications (MF2004 edition) edited to reflect the scope of work of the project.

## B. ARCHITECTURAL

1. Submit a 100% complete and coordinated set of construction drawings and specifications.
2. Submit fully dimensioned floor plans showing all revisions required by comments from the previous review meetings.
3. Submit interior details.
4. Submit drafting symbols, and abbreviations, general notes and schedules that are complete and coordinated with all contract documents.
5. Submit a complete and coordinated finish schedule.
6. Submit demolition plans, existing finish schedule and notes, on demolition plans.
7. Submit completed building sections, wall sections and exterior elevations that show finish floor elevations, and indicate all building systems and materials.
8. Submit completed reflected ceiling plans for entire buildings, indicating all ceiling mounted equipment, lighting fixtures, air diffusers, registers, etc.
9. Submit equipment plans and details with all revisions from comments on previous submittal.

10. Provide a list of provision of LEED meeting the VA Energy Conservation requirements.

#### C. INTERIOR DESIGN

NA

#### D. ARCHITECTURAL SPECIFICATIONS

1. Assure the original specification drafts have been edited and tailored in their application to represent accurate coordination between the drawings and specifications.
2. Submit one full set of drawings of all disciplines, fully coordinate.
3. Submit a brief description of work for inclusion in the Pre-Solicitation Notice.
4. Final Bid Document Submittal:
  - a. Revise draft specifications to incorporate:
    1. All changes, resolution of conflicts and modifications noted in previous reviews.
    2. Results of any drawing changes not shown on the drawings that affect the specifications.
  - b. Type the specifications in final format and content including any desk copy changes made by the VA staff at the previous reviews. Submit

a complete set of the typed architectural specifications including one full set of final drawings of all disciplines, fully coordinate.

#### E. EQUIPMENT SPECIFICATIONS

NA

#### F. STRUCTURAL

1. Submit applicable sections of the VA Master Specifications (MF2004 edition) edited to reflect the scope of work of the project.
2. Drawings and edited specification documents shall not contradict each other.
3. Submit all calculations.
4. Show necessary typical details correlating architectural and mechanical features.
5. Show sections and details so that construction features can be defined, including the interfacing with new to existing construction.

#### G. SANITARY

1. Submit 100% complete drawings to include the following:
  - a. Previously submitted drawings that have incorporated comments of the last review.
  - b. Legend, notes, and details.

#### H. PLUMBING



1. Submit applicable sections of the VA Master Specifications (MF2004 edition) edited to reflect the scope of work of the project.
2. Drawings and edited specification documents shall not contradict each other.
3. Submit all calculations.
4. Provide complete selection and sizing data, including catalog cuts and calculations.
5. Submit any applicable equipment schedules.

#### I. HEATING, VENTILATION AND AIR CONDITIONING (HVAC)

NA

#### J. ASBESTOS/PCB ABATEMENT

1. Provide details, sections, notes and other information that is necessary for complete bid documents.
2. Prepare project specifications by editing and modifying VA Master Specifications on asbestos and PCB abatements.
3. Provide the drawings and specifications for the restoration of the impacted building sub-systems as an integral part of the overall project design. Provide for replacing of removed asbestos insulation with acceptable non-asbestos insulation, replacing contaminated carpets, replacing old contaminated light fixtures and replacing partially torn down plaster ceilings. Coordinate work with the Architectural/Engineering Firm.
4. Provide an integrated phasing plan showing how the abatement and the general modification work will be executed. Provide some reasonable

cushion between each abatement phase and the following general construction phase to allow for unforeseen delays in abatement completion.

5. Provide a major and minor summary of the square feet of floor space for all abatement areas, the total linear and square feet of asbestos to be abated and the total cost of abatement. Include in the cost estimate any cost for decontamination of equipment and fixtures. Provide a copy of the above summary to the construction cost estimator for inclusion as a separate bid item in the project estimate.
6. Perform Task 5 according to the "Special Instructions – Asbestos/PCB Abatement" document (attached)

#### K.LEAD ABATEMENT

1. Provide details, sections, notes and other information that is necessary for complete bid documents.
2. Prepare project specifications by editing and modifying VA Master Specifications on LEAD abatement.
3. Provide the drawings and specifications for the restoration of the impacted building sub-systems as an integral part of the overall project design. Provide for replacing of removed LEAD with acceptable non-asbestos insulation, replacing contaminated carpets, replacing old contaminated light fixtures and replacing partially torn down plaster ceilings. Coordinate work with the Architectural/Engineering Firm.

4. Provide an integrated phasing plan showing how the abatement and the general modification work will be executed. Provide some reasonable cushion between each abatement phase and the following general construction phase to allow for unforeseen delays in abatement completion.
5. Provide a major and minor summary of the square feet of floor space for all abatement areas, the total linear and square feet of LEAD to be abated and the total cost of abatement. Include in the cost estimate any cost for decontamination of equipment and fixtures. Provide a copy of the above summary to the construction cost estimator for inclusion as a separate bid item in the project estimate.
6. Perform Task 5 according to the "Special Instructions - LEAD Abatement" document (attached)

#### L. ELECTRICAL

1. Submit applicable sections of the VA Master Specifications (MF2004 edition) edited to reflect the scope of work of the project.
2. Drawings and edited specification documents shall not contradict each other.
3. Submit all calculations
4. Provide complete selection and sizing data, including catalog cuts and calculations
5. Submit any applicable equipment schedules
6. Provide a sheet for symbols, notes and abbreviations used at this submission.
7. Submit a full set of electrical, lighting, power, and signal floor plans.
8. Provide demolition plans to indicate the complete electrical work in all areas to be renovated.

9. Phasing and shutdown requirements.

#### M. BARRIER FREE DESIGN

1. Complete all drawings and specifications. Ensure that every effort to make this facility accessible to handicapped employees and patients has been taken in this design.

#### N. FIRE PROTECTION

1. Complete all drawings and specifications. Ensure compliance with NFPA. Provide hydraulic calculations used to derive the pipe and equipment dimensions and ratings.

#### O. PHASING REQUIREMENTS

1. Phasing requirements shall describe the general sequence of the project work, estimated project duration (including allowances for delivery items), and what government constraints will exist that will influence the Contractor's approach to the construction project. In addition, special attention shall be given to asbestos abatement requirements to ensure that the project phasing plan and associated cost are reasonable. Adequate time shall be allocated for the evaluation for asbestos abatement areas in addition to the time to perform the asbestos abatement work that must precede the general construction. The Architectural/Engineering Firm shall submit the following phasing information:

- a. Phasing Narrative in written form which outlines phasing requirements and sequence with all areas of the project identified as a part of some phase. Each phase description shall include constraints particular to that phase, what other phases that must precede it, and any VA moves which must precede the start of the phase or phases. If equipment and other removable items require storage and relocation by the government, because of asbestos abatement, these requirements shall be listed in the phasing narrative. Special phasing constraints which may be common to the project should be listed at the end of the narrative and not within each individual phase description.
- b. Individual phases shall be outlined and labeled on all drawings including site, architectural, Structural, plumbing and electrical drawings. Phases shall be outlined on the submitted full sized drawings.
- c. All systems shall be designed so that, upon completion of a particular phase, the entire area covered by that phase can be occupied by the Medical Center personnel with all systems functioning properly.

#### P. Estimating

1. Engineering firm will provide a complete and accurate cost estimate.
2. Construction estimate will be in R.S. Means format

AE shall submit two (2) full size sets and one (1) half size set of hard copies design drawings and specifications including an electronic copy in PDF format. Also, all electronic

copies including CAD format drawings will be provided to the VA via mandated Autodesk Build File Management system.

AE shall also submit a copy of comment resolution addressing fire protection issues to the VA for review.

## CONTRACT DOCUMENTS

1. The Architectural/Engineering Firm shall deliver the original contract drawings to the Project Manager for signature, after the previous review comments are incorporated and the contract drawings have been approved. The contract drawings shall bear the signature and seal of the Registered Architect and Professional Engineer responsible for the design.
  - a) 3 sets of printed Contract drawings. (All drawings stamped and signed)
  - b) Provide electronic copy of drawings in AutoCAD most current version transmitted via Auto Desk Build (Electronic File Management system).
  - c) Provide electronic copy of drawings in PDF format transmitted via Auto Desk Build (Electronic File Management system).
2. Submit a complete set of applicable VA Master Specifications edited to reflect the scope of work of the project. Also, submit copies of any architectural specifications prepared by the Engineering Firm from any source other than VA Master Specifications. Ensure the specifications have been edited and tailored in their application to represent accurate coordination between the drawings and specifications.

- a) Provide electronic copy of specifications in PDF and WORD format transmitted via Auto Desk Build (Electronic File Management system).
- b) Provide bound and stamped hardcopy of specifications.
- c) AE site visit for Pre-Bid conference.
- d) Answer questions from prospective bidders about the project design.
- e) AE provide Submittal Log listing the anticipated required submissions and indicated receipt (critical) priority.
- f) AE provide the final catalog cut sheets and manufacturers literature for all equipment.

## **SOLICITATION SUPPORT SERVICES**

Pre-bid site visit to include meeting and project walk around. This is the meeting where contractors are invited to the job site to visually inspect the construction site to accurately develop a construction bid package.

Answer any technical questions/requests for information (RFIs) received from contractors over the duration of the solicitation, including questions asked during the site visit. Questions from contractors may come at multiple times and cover several questions over the course of the solicitation period. Technical question responses will be provided to the Contracting Officer who will distribute the information via amendment to the solicitation.

Answers to RFIs will be required from the A-E within five calendar days of the date the questions are sent to the A-E from either the project COR or CO. An alternate response period may be submitted for review, however, unless approved by the CO and COR, the A-E is required to comply with the five-calendar-day requirement.

Throughout the solicitation period, the A-E shall provide addenda as necessary if required to revise specs and drawings.

In response to any technical questions and/or addenda issued during the solicitation period that require edits to the issued specifications and/or drawings, the A-E shall provide any final revisions to the 100% Construction Documents as part of the construction contract. The A-E shall make these changes within 14 calendar days of closing of the solicitation. These documents shall be labeled Final Construction.

## CONSTRUCTION PERIOD SERVICE

1. Assist in the development of shop drawings, and review submittals for acceptance.

(1) The A/E shall check Government furnished and/or the Construction Contractor's shop drawings, detail drawings, schedules, descriptive literature and samples, testing laboratory reports, field test data and review the color, texture and suitability of materials for conformity with the design concept and construction documents.

(2) The A/E shall recommend approval, disapproval, or other suitable disposition. He shall evaluate the submittals with reference to any companion submittals that constitute a system. When necessary, the A/E will, through the COR, request the Contractor to submit related components of a system before acting on a single component. Should this procedure be inappropriate, the A/E shall review all prior submittals for related components of the system before acting on a single component. The A/E may be required to hold joint reviews with the VA technical staff on complicated system submittals. The A/E shall notify the COR in writing of all deviations from the requirements of the construction documents that he has found in the Construction Contractor's submittals. Such responsibility shall be carried out with reasonable



promptness to cause no delay.

(3) The A/E shall submit recommendations on all shop drawings and other submittals and other requests within fifteen (15) calendar days after receipt.

2. The A/E, when requested, shall assist the Contracting Officer and his COR to interpret the construction documents and shall:

(1) recommend any action(s) he deems suitable for the satisfactory prosecution of the construction work.

(2) prepare any supplemental drawings, specifications or other documents that may be required to clarify or supplement the construction documents.

(3) assess the preparation of Construction Contract Modifications initiated by or through the VA to be executed in accordance with the Construction Documents.

Modification requests shall be submitted within seven (7) calendar days of submittal to the A/E. This includes the review of the contractor's change order/modification proposal for construction change orders, as well as the formulation of an IGCE for the modification, as requested. The A/E's reply regarding his/her review of cost proposals and his/her recommendations shall include an independent breakdown of costs in detail, quantities, and unit prices and shall cover both additions and deductions of labor, materials, and equipment. The A/E shall also supply cost estimates as requested by the COR for Request for Proposals for anticipated changes." All documents

in any of the electronic formats listed above. Note - If major modification, VA will provide additional time and document format guidance.

3. Record drawings shall be delivered to the VA within thirty (30) calendar days after VA acceptance of the construction project. Autodesk Revit Project (.rvt) format and AutoCAD (.dwg) format.
  - (a) The COR shall furnish to the A/E a red line marked-up set of prints showing all changes made during the construction period. The A/E shall field verify and incorporate into the drawings all significant changes marked-up by the COR and, any significant changes made by solicitation amendments and construction contract modifications. The revision portion of title block shall be dated and noted "Record Drawings". The A/E shall prepare record drawings by making changes on the original contract drawings. A copy of the up-dated original will be sent to the COR for verification. Upon the verification the A/E will submit the originals on CD-ROM.
  - (b) The A/E shall deliver the record drawings to the COR within sixty (60) days after receiving the marked-up documents from the COTR. Submission shall be subject to the approval of the COR as to quality and completeness.
4. Conduct periodic inspection services. These "site visits" also include a pre-construction conference, intermediate and a final inspection visit. The Engineering Firm will submit a survey report summarizing the conditions found at the construction site as a result of the site visit.
  - (a) Intermediate Inspections. The A/E shall provide registered architects and engineers who are familiar with the design work of the project to act as

observers and advisers to the VA team conducting intermediate inspections.

(b) Partial Final and Final Inspections. The A/E shall provide registered architects and engineers in the number and of the disciplines at the time requested by the Contracting Officer to act as observers and advisers with a VA team conducting partial final and/or final inspection(s) of the project.

(c) The A/E shall conduct Final inspection submit a report of actual work accomplished for purpose of acceptance & punch list, and issue report within five (5) calendar days of the site visit.

5. Provide detailed itemized cost estimates and aid in the development of supplemental agreements amendments to the contract (if necessary). Perform change order reviews and provide solutions as needed.

## ASBESTOS/PCB TASKS 1-9

### TASK 1:

Inspect project areas suspected of containing asbestos and PCB. The CIH shall submit a preliminary description of asbestos and PCB assessments and abatement problems and solutions. This preliminary report shall include the following:

- a. Summary results of review of building records.
- b. Summary results of interview of station personnel.
- c. Determination of whether any materials used in the building construction are known to contain asbestos and PCB, existing asbestos survey are to be consulted. Some trade name products are known to contain asbestos, these must be identified.
- d. Results of visual inspection of the building to determine location and condition of asbestos materials.
- e. Presentation of sampling strategy that will yield statistically a viable conclusion on the extent of asbestos and PCB present.

### TASK 2:

Sample suspect materials and select a qualified laboratory to perform analysis. Provide the name and location of a qualified laboratory for sample analysis that the CIH intends to utilize. The following materials shall be tested for PCB, but not to be limited to window caulking, transformers, switch thermostats.

### TASK 3:

Review sample analysis and determine the most appropriate approach to take during construction. At this time the CIH should have collected appropriate samples, selected a qualified laboratory for sample analysis, evaluated sample analysis results to determine adequacy for an assessment report, and ensured that the sample analysis results and number of samples taken are adequate to provide the CIH with confidence that the maximum detection of asbestos and PCB has been achieved.

### TASK 4:

Submit laboratory analysis and recommended actions. Submit an exposure assessment report on the impacted areas of the project. Include the following information:

- a. Describe the impacted areas by sketches, room names or other means. Show or describe the location of each sample taken, its composition, and identification number.
- b. Classify each impacted space as "Major Abatement", "Minor Abatement", or "No Abatement" areas. A "Major Abatement" area requires full protective measures - sealing off, change room, shower room, etc; and removal of all asbestos within the limits of the area. A "Minor Abatement" area can be decontaminated with limited applications of the "Glove and Bag" approach. A "No Abatement" area requires no abatement work.
- c. For each "Major Abatement" area, identify area location within building and provide quantity and cost estimates of all asbestos and PCB within the limits of the area. Quantities are to be in linear feet of pipe and square feet of ceilings, walls and floor materials. The cost is to be derived from the quantities multiplied by a unit price.
- d. For each "Minor Abatement" area, provide the description, size, location of each pipe element to be decontaminated, and a cost estimate.
- e. Provide a summary of the square feet of floor space for all major and minor abatement areas, the total linear and square feet of asbestos and PCB to be abated, the total cost of abatement and a preliminary schedule of completion. Include in the cost estimate any cost for decontamination of equipment and fixtures. Provide a copy of the above summary to the construction cost estimator for inclusion as a bid item in the project estimate. Work will be done in phases. Consult with the VA on this so that phasing is reflected in cost.
- f. Provide an assessment on the impact of abatement on building sub-systems such as ceilings, lights, walls, carpets, pipes and equipment. Provide room-by-room description and estimated quantities of impact items to be replaced (number of lighting fixtures, square feet of ceiling, etc.), and a summary of these quantities. Provide a cost estimate for all impacted items.
- g. Provide a signed cover statement; this statement shall be in accordance with the "Sample Cover Statement" provided by the VA as a part of the A/E package.
- h. Evaluate the removal cost versus the deferral of some asbestos removal. The VA will furnish a CIH with a list of areas asbestos would be abandoned in place. This CIH will then review and make recommendations, and incorporate a final decision into the design.

#### TASK 5:

Design and specify the appropriate techniques and approaches for the execution of the recommended actions in TASK 4, and submit specifications and drawings. Verify variances from New York State Code Rule 56.

Prepare drawings and specifications for the asbestos abatement work reflecting the assessment report and paragraph 4(h). The drawings shall be separately identified from the rest of the drawings, could be 1/16" scale plans (if convenient), and shall provide as a minimum the following information:

- a) For each "Major Decontamination" area, show the limits of containment, the location and quantities of asbestos material, the arrangements for auxiliary rooms (such as change room, shower room, etc.), the engineering of the negative air systems, the path of asbestos waste transportation and the location and connection to required utilities.
- b) For each "Minor Decontamination" area, show on plans the exact location type and length of pipe element to be abated by the "Glove and Bag" approach, and any other abatement features. CIH shall propose the proper type of PPE for each abatement.
- c) Provide a major and minor summary of the square feet of floor space for all abatement areas, the total linear and square feet of asbestos to be abated and the total cost of abatement. Include in the cost estimate any cost for decontamination of equipment and fixtures. Provide a copy of the above summary to the construction cost estimator for inclusion as a separate bid item in the project estimate.
- d) Provide details, sections, notes, and other information that is necessary for complete bid documents.
- e) Prepare project specifications by editing and modifying VA master specifications on asbestos abatements.
- f) Provide information and data for the restoration of the impacted building sub-systems as an integral part of the overall project design. Provide information and data for replacing the removed asbestos insulation with acceptable non-asbestos insulation. Replace old contaminated light fixtures and replace partially torn down plaster ceilings.
- g) Incorporate comment from the VA Hudson Valley and/or VISN Industrial Hygienist. This review is to ensure compliance with VA regulations and policies.
- h) Provide VA Engineering with an itemized estimate for Tasks 6 through Task 9 received as part of Task 2.

#### TASK 6:

Assist VA in evaluating qualifications of contractor and personnel by establishing minimum qualification requirements for specialization experience, key personnel, standard operating procedures, etc.

#### TASK 7:

Continuously monitor performance of contractor during abatement work to ensure adherence to abatement design. Monitor air quality in and around work space, worker exposure, operating procedures, respiratory protection systems, the abatement process, and the packing, transportation, and disposal of asbestos and PCB. Secure the services of a qualified analytical laboratory to turn over sample results within 24 hours unless a specific request has been made by the VA.

#### TASK 8:

At the end of abatement and after final cleaning, inspect the workspace and perform the required testing to establish the decontamination level specified.

#### TASK 9:

Provide a certificate to the Resident Engineer certifying that the abatement process was performed in accordance with the best practice, the maximum feasible protection of people and the environment has been achieved during the abatement process, and the impacted space has achieved the VA required decontamination levels.

All data, inspection results and testing results generated by the CIH will be available to the contractor for information and consideration.

Monitoring and inspection results of the CIH will be used by the VA to issue any stop work or removal orders to the contractor during abatement work and to accept or reject the area or building as decontaminated.

The CIH will make available to the contractor the plan for sample collection and analysis for continuous monitoring outside the work area and the plan for final inspection and testing for each space or building prior to executing each plan. The plan will include the locations for samples, name and qualifications of person taking samples, the methodology of analysis and the name and address of the laboratory to be used for analysis.

LEAD –LBP CIH TASKS 1-9

#### TASK 1:

Inspect project areas suspected of containing Lead and Lead Based Paint (LBP). CIH shall submit a preliminary description of assessment and abatement problems and solutions. This preliminary report shall include the following:

- a. Summary results of review of building records.
- b. Summary results of interview of station personnel.
- c. Determination of whether any materials used in the building construction are known to contain lead. Some trade name materials are known to contain lead and should be identified.
- d. Results of visual inspection of the building to determine location and condition of lead materials.
- e. Presentation of sampling strategy that will yield statistically viable conclusion on the extent of asbestos present.

#### TASK 2:

Sample suspect materials and select a qualified laboratory to perform analysis. Provide the name and location of a qualified laboratory for sample analysis which the CIH intends to utilize.

#### TASK 3:

Review sample analysis and determine the most appropriate approach to take during construction. At this time the CIH should have collected appropriate samples, selected a qualified laboratory for sample analysis, evaluated sample analysis results to determine adequacy for an assessment report, and ensured that the sample analysis results and number of samples taken are adequate to provide the CIH with confidence that the maximum detection of lead has been achieved.

#### TASK 4:

Submit laboratory analysis and recommended actions. Submit an exposure assessment report on the impacted areas of the project. Include the following information:

- a. Describe the impacted areas by sketches, room names or other means. Show or describe the location of each sample taken, its composition, and identification number.
- b. Classify each impacted space as area requires full protective measures - sealing off, change room, shower room, etc; and removal of all within the limits of the area or approach "No Abatement" area requires no abatement work and encapsulation with sealers.
- c. For each Abatement area, identify area location within building and provide quantity and cost



estimates of all lead within the limits of the area. Quantities are to be in linear feet and square feet of ceilings and walls. The cost is to be derived from the quantities multiplied by a unit price.

d. Include in the cost estimate any cost for decontamination of equipment and fixtures. Provide a copy of the above summary to the construction cost estimator for inclusion as a bid item in the project estimate. Work will be done in phases. Consult with the VA on this so that phasing is reflected in cost.

f. Provide an assessment on the impact of abatement on building sub-systems such as ceilings, lights, walls, carpets, pipes and equipment. Provide room by room description and estimated quantities of impact items to be replaced (number of lighting fixtures, square feet of ceiling, etc.), and a summary of these quantities. Provide a cost estimate for all impacted items.

g. Provide a signed cover statement. This statement shall be in accordance with the "Sample Cover Statement" provided by the VA as a part of the A/E pack Autodesk Architectural Desktop.

h. Evaluate the removal cost versus the deferral of some removal. The VA will furnish a CIH with a list of areas lead would be abandoned in place. This CIH will then review and make recommendations, and incorporate a final decision into the design.

#### TASK 5:

Design and specify the appropriate techniques and approaches for the execution of the recommended actions in TASK 4, and submit specifications and drawings. Verify if the State Specification will have to be adhered to.

Prepare drawings and specifications for the abatement and corrective work reflecting the assessment report and paragraph 4. The drawings shall be separately identified from the rest of the drawings, could be 1/16" scale plans (if convenient), and shall provide as a minimum the following information:

a. For each "Decontamination" area show the limits of sealing off, the location and quantities of material, the arrangements for auxiliary rooms (such as change room, shower room, etc.), the engineering of the negative air systems, the path of debris transportation to the loading platform, and the location and connection to required utilities.

b. Provide a summary of the square feet of floor space for all abatement areas, the total linear and square feet of asbestos to be abated and the total cost of abatement. Include in the cost estimate any cost for decontamination of equipment and fixtures. Provide a copy of the above summary to the construction cost estimator for inclusion as a separate bid item in the project estimate.

c. Provide details, sections, notes, and other information that is necessary for complete bid documents.

- d. Prepare project specifications by editing and modifying VA master specifications on LBP abatements.
- e. Provide information and data for the restoration of the impacted building sub-systems as an integral part of the overall project design. Provide information and data for replacing old contaminated light fixtures and replace partially torn down plaster ceilings.
- f. Incorporate comment from the Department of Veterans Affairs, Region 1, and Industrial Hygienist. This review is to ensure that VA regulations and policies are adhered to.
- g. Provide VA Engineering with an itemized estimate for Tasks 6-9 received as part of Task 2.

#### TASK 6:

Assist VA in evaluating qualifications of contractor and personnel by establishing minimum qualification requirements for specialization experience, key personnel, standard operating procedures, etc.

#### TASK 7:

Continuously monitor performance of contractor during abatement work to ensure adherence to abatement design. Monitor air quality in and around work space, worker exposure, operating procedures, respiratory protection systems, the abatement process, and the packing, transportation, and disposal of debris following the test analysis to determine the classification of disposal. Secure the services of a qualified analytical laboratory, to turn over sample results within 24 hours.

#### TASK 8:

At the end of abatement and after final cleaning, inspect the work space and perform the required testing to establish the decontamination level specified.

#### TASK 9:

Provide a certificate to the Resident Engineer certifying that the abatement process was performed in accordance with the best practice, the maximum feasible protection of people and the environment has been achieved during the abatement process, and the impacted space has achieved the VA required decontamination levels. Review the abatement contractor close out documents, collect manifests for disposal and submit package to VA. (2- sets).

END